

Bhanu Sri Vaishnavi Bharatula

Atlanta, GA, USA

30338

9044301808

bbsvaishnavi@gmail.com

<https://www.linkedin.com/in/bbsvaishnavi/>

23-Oct-2025

Dear Hiring Committee,

I am excited to apply for the Software Developer position within the Robotics Department at Michigan Engineering. With a Master's in Computer Science (Georgia State University) and extensive experience in full-stack development, Python, and JavaScript, I am well-prepared to contribute to the department's mission of advancing robotics research and educational initiatives.

In my previous roles, I have:

- **Developed and maintained full-stack applications** using C#, .NET Core, Angular, and SQL databases, improving system performance and user experience for thousands of active users.
- **Implemented scalable solutions in cloud environments** (AWS, Azure) and applied CI/CD pipelines to ensure high-quality deliverables.
- **Applied data analysis and machine learning** in research settings, including neuroimaging studies with Python, R, and MATLAB, supporting large datasets and generating actionable insights.
- **Collaborated with non-technical stakeholders** to gather requirements and translate them into practical, efficient software tools.
- **Ensured software quality and security**, writing unit tests, monitoring applications, and adhering to best practices for compliance and operational integrity.

I am particularly drawn to Michigan Engineering's emphasis on innovation, inclusivity, and societal impact. My experience in higher-education research and software development positions me to effectively support the Robotics Department's applications, streamline workflows, and contribute to collaborative projects.

I would welcome the opportunity to discuss how my technical expertise and research experience align with your team's goals. Thank you for considering my application.

Sincerely,

Bhanu Sri Vaishnavi Bharatula

Bhanu Sri Vaishnavi Bharatula

+1 (904) 430-1808 | bbsvaishnavi@gmail.com | <https://linkedin.com/in/bbsvaishnavi/>

SKILLS

Languages: Python, C#, C++, C, Java, SQL, Visual Basic

Web Development: ASP.NET (MVC & Core), .NET Core, Angular, React, Node.js, HTML, CSS, JavaScript, TypeScript, PHP, Bootstrap, jQuery

Databases: MS SQL Server, SQLite, PostgreSQL

Tools & Methodologies: Azure DevOps, GitHub, Visual Studio, VS Code, Jira, Agile, Unit Testing, Microsoft Excel

Devops & Cloud: AWS, Docker

EDUCATION

Georgia State University

Master of Science, Computer Science

August 2024 - December 2025

GPA: 4

VR Siddhartha Engineering College

Bachelor's in Technology, Information Technology

June 2018 - May 2022

GPA: 3.7

PROFESSIONAL EXPERIENCE

Research Assistant (Machine Learning)

January 2025 – Present

TReNDS Lab - Georgia State University

- Conducted neuroimaging and behavioral research using Python, MATLAB, and R to investigate brain connectivity and cognitive function across >11,000 subjects.
- Applied machine learning models to uncover significant neural-behavioral correlations, identifying key predictive biomarkers of cognitive and emotional traits.
- Utilized ICA decomposition to localize brain regions associated with behavioral traits such as anxiety and depression. Quantified effect sizes, cross-validated model performance, and assessed statistical significance to ensure robust and reproducible results.

Software Engineer

August 2022 – August 2024

Deloitte

- Developed full-stack applications using .NET (C#) and Visual Basic (VB) for backend, Angular for frontend, and MS-SQL for database management, improving system performance by 15% and enhancing user experience for 2,000+ active users.
- Implemented REST APIs and Redis for distributed caching and session state management in the Incentives Hub application, supporting seamless data exchange across 5+ internal services.
- Engineered scalable solutions for complex tax requirements in Azure and AWS environments and achieving 95% unit test coverage through efficient CI/CD pipelines within Agile SDLC.

Full Stack Developer

March 2022 – July 2022

A Plus Topper

- Refined an academic quiz web application by developing a full-stack solution with Angular (frontend), Django (backend), and SQL (database), reducing page load time by 10% and improving user experience for 500+ users.
- Authored a data automation tool using Django and Selenium web scraping, reducing data retrieval time from hours to 5–10 minutes and streamlining content acquisition workflows by 70%.
- Implemented authentication, form validations, and API integrations in Angular and Django REST Framework, enhancing application security and usability.

Software Engineer Intern

June 2021 – July 2021

Deloitte

- Developed a .NET-based web application integrated with a Random Forest model to predict future traffic volumes with 95% accuracy, enabling proactive traffic management.
- Enhanced user experience by implementing advanced search functionality and secure authentication, including email verification, reducing unauthorized access by 100%.

PROJECTS

Hyperspectral Image Classification using Deep Learning

- Led the development of a robust system for hyperspectral image classification, leveraging 2D CNNs and machine learning models (SVM, Random Forest, and Ensemble) to analyze agricultural datasets.
- Achieved 99% accuracy utilizing remote sensing data to accurately classify crops and land features.

Virtual Classroom Assistant

- Developed a web platform integrating video conferencing, automated speech-to-text transcription, and report generation using Django, Jitsi Meet API, and SQLite, improving remote learning accessibility.
- Engineered a scalable transcription solution achieving 90% accuracy, streamlining report generation and enhancing collaboration for educators and students.